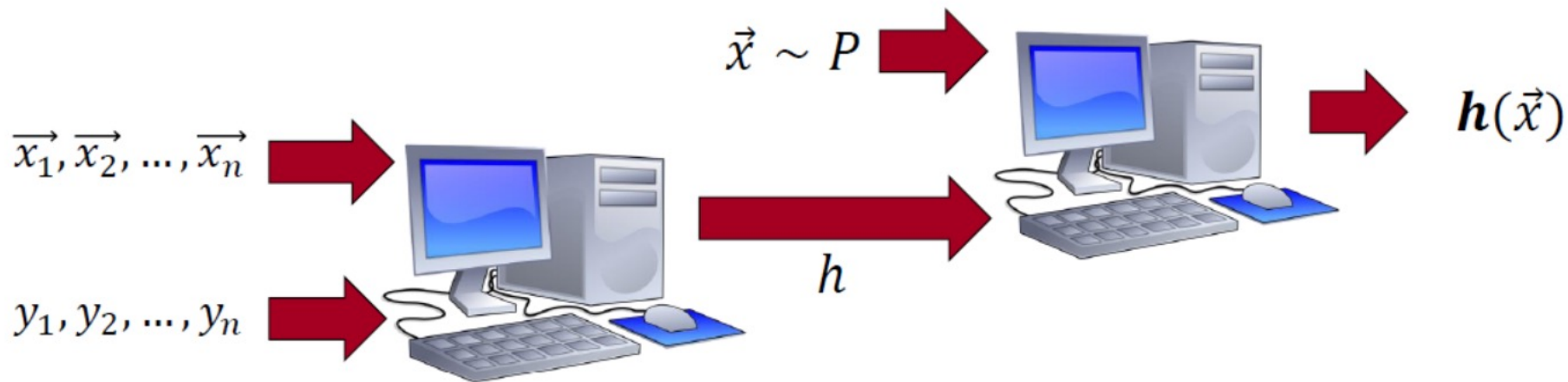


LECTURE 8

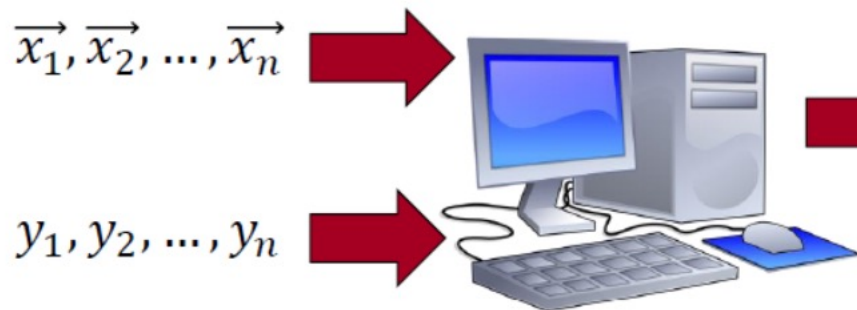
What is Machine Learning?

Training

Testing



Training



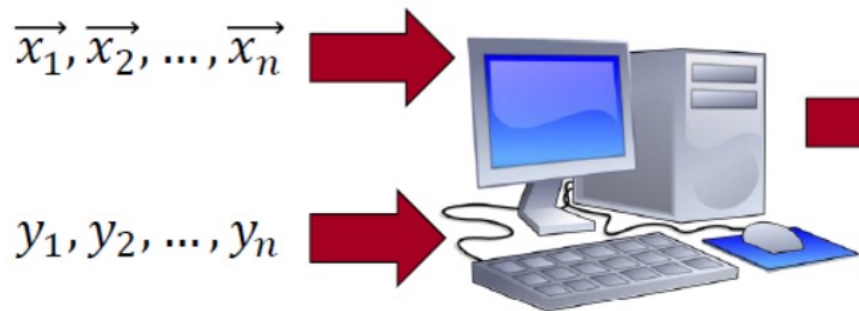
Testing



Evaluation

Compare $h(\vec{x})$ and y

Training



Testing



Evaluation

Compare $h(\vec{x})$ and y

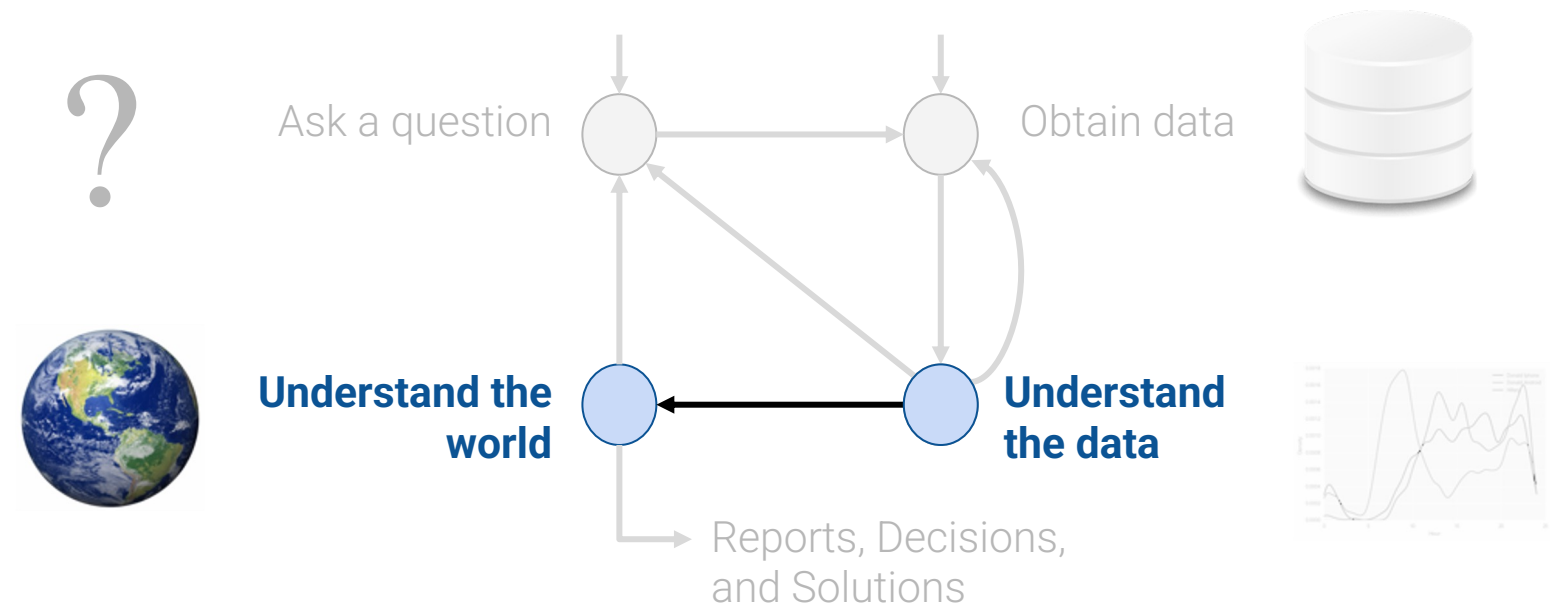
Our hope: $h(\vec{x}) \approx y$

Week 8

Introduction to Modeling, SLR

Understanding the usefulness of models and the simple linear regression model

Plan for Next Few Lectures: modeling



(today)

Modeling I:
Intro to Modeling, Simple
Linear Regression

Modeling II:
Different models, loss
functions, linearization

Modeling III:
Multiple Linear
Regression

Today's Roadmap

- What is a Model?
 - Regression Line, Correlation
- The Modeling Process
 - Choose a Model
 - Choose a Loss Function
 - Fit the Model
 - Evaluate the Model

What is a Model?

- **What is a Model?**
 - Regression Line, Correlation
- The Modeling Process
 - Choose a Model
 - Choose a Loss Function
 - Fit the Model
 - Evaluate the Model

What is a Model?

A model is an **idealized representation** of a system.

Example:

We model the fall of an object on Earth as subject to a constant acceleration of 9.81 m/s^2 due to gravity.

- While this describes the behavior of our system, it is merely an approximation.
- It doesn't account for the effects of air resistance, local variations in gravity, etc.
- But in practice, it's accurate enough to be useful!

Essentially, all models are wrong, but some are useful.



George Box, Statistician
(1919-2013)

Known for “All models are wrong”
Response-surface methodology
EVOP
q-exponential distribution
Box–Jenkins method
Box–Cox transformation

Three Reasons for Building Models

Reason 1:

To explain **complex phenomena** occurring in the world we live in.

- How are the parents' average heights related to the children's average heights?
- How do an object's velocity and acceleration impact how far it travels?

Often times, we care about creating models that are **simple and interpretable**, allowing us to understand what the relationships between our variables are.

Reason 2:

To make **accurate predictions** about unseen data.

- Can we predict if an email is spam or not?
- Can we generate a one-sentence summary of this 10-page long article?

Other times, we care more about making extremely accurate predictions, at the cost of having an uninterpretable model. These are sometimes called **black-box models**, and are common in fields like deep learning.

Reason 3:

To make **causal inferences** about if one thing causes another thing.

- Can we conclude that smoking *causes* lung cancer?
- Does a job training program cause increases in employment and wage?

Much harder question because most statistical tools are designed to infer association not causation

This won't be the focus of this class, but will be if you go on to take more advanced classes (Stat 156, Data 102)

Most of the time, we want to strike a balance between **interpretability** and **accuracy**.

Common Types of Models

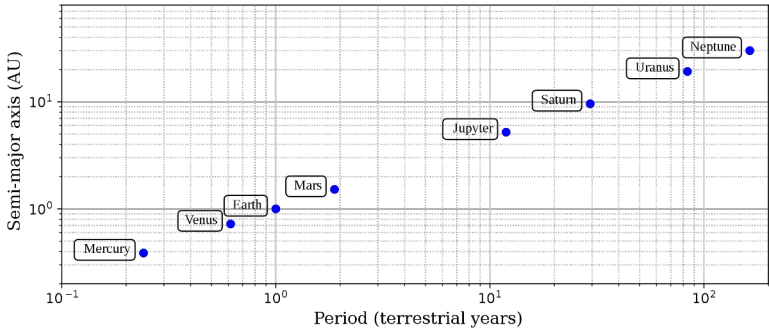
Deterministic physical (mechanistic) models: Laws that govern how the world works.

[Kepler's Third Law of Planetary Motion](#) (1619)

The ratio of the square of an object's orbital period with the cube of the semi-major axis of its orbit is the same for all objects orbiting the same primary.

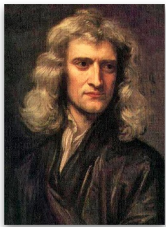


$$T^2 \propto R^3$$



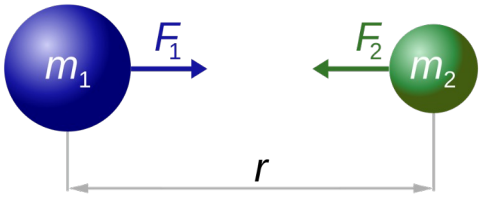
[Newton's Laws: motion and gravitation](#) (1687)

Newton's second law of motion models the relationship between the mass of an object and the force required to accelerate it.

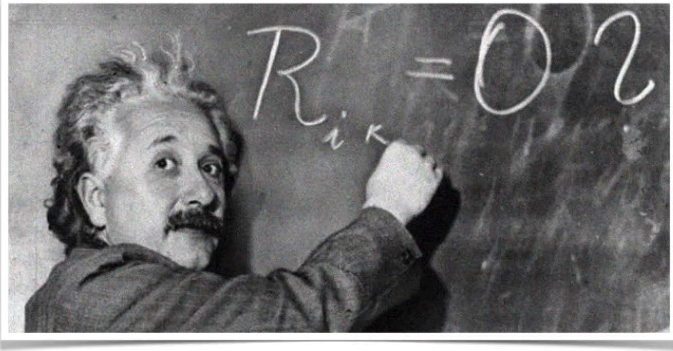


$$\mathbf{F} = m\mathbf{a}$$

$$F = G \frac{m_1 m_2}{r^2}$$

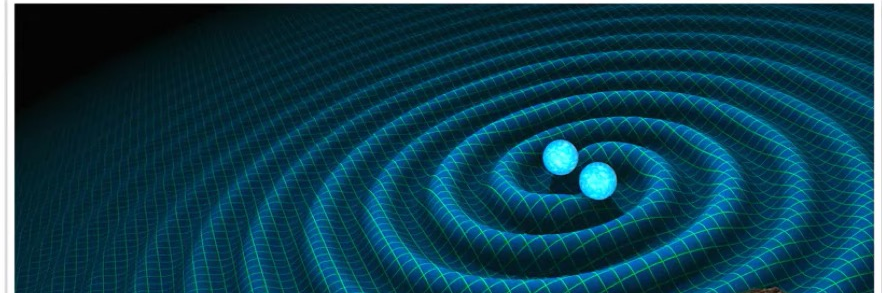


A long time ago in a galaxy far, far away...



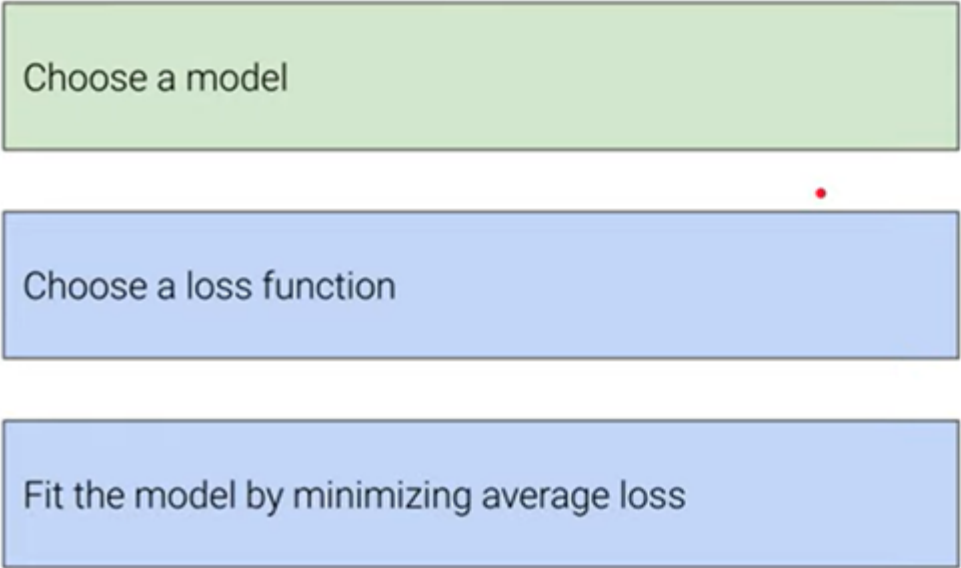
$$R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Einstein's Field Equations of General
Relativity
Annalen der Physik, 1916



The modeling process

We've implicitly introduced this three-step process, which we will use constantly throughout the rest of the course.



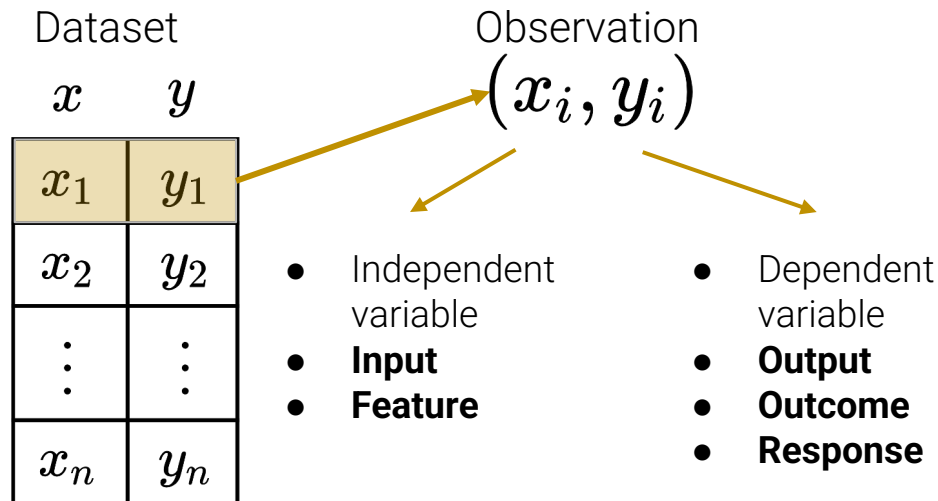
Choose a model

Choose a loss function

Fit the model by minimizing average loss

Models in Data Science

we'll treat a model as some mathematical rule to describe the relationships between variables.



Prediction

If we use x to predict y , the predictions are denoted as $\hat{y} = \{\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n\}$

Models

Some models we will see in the next few lectures:

$$\hat{y}_i = \theta_0 + \theta_1 x_i$$
$$\hat{y}_i = \theta$$
$$\hat{y}_i = x_i^\top \theta$$

Parametric models

Models in Data Science: Parametric Models

Parametric models are described by a few **parameters** ($\theta_0, \theta_1, \theta$, etc.)

- No one tells us the parameters: the data informs us about them.
- The x, y values are **not** parameters because we directly observe them.
- Sample-based **estimate** of parameter θ is written as $\hat{\theta}$.
- Usually, we pick the parameters that appear "**best**" according to some criterion we choose

θ	Model parameter(s)	$\left\{ \begin{array}{l} \hat{y} = \theta_0 + \theta_1 x \end{array} \right.$	Any linear model with parameters $\theta = [\theta_0, \theta_1]$
$\hat{\theta}$	Estimated parameter(s), "best" fit to data in some sense	$\left\{ \begin{array}{l} \hat{y} = \hat{\theta}_0 + \hat{\theta}_1 x \end{array} \right.$	The "best" fitting linear model with parameters $\hat{\theta} = [\hat{\theta}_0, \hat{\theta}_1]$

Models in Data Science: Parametric Models

Parametric models are described by a few **parameters** ($\theta_0, \theta_1, \theta$, etc.)

- No one tells us the parameters: the data informs us about them.
- The x, y values are **not** parameters because we directly observe them.
- Sample-based **estimate** of parameter θ is written as $\hat{\theta}$.
- Usually, we pick the parameters that appear "**best**" according to some criterion we choose

Note: Not all statistical models have parameters!

KDEs, k-Nearest Neighbor classifiers are non-parametric models.

θ Model parameter(s) linear model with parameters $\theta = [\theta_0, \theta_1]$

$\hat{\theta}$ Estimated parameter(s), "best" fit to data in some sense $\hat{y} = \hat{\theta}_0 + \hat{\theta}_1 x$ The "best" fitting linear model with parameters $\hat{\theta} = [\hat{\theta}_0, \hat{\theta}_1]$

The Modeling Process

1. Choose a model

How should we represent the world?

2. Choose a loss function

How do we quantify prediction error?

3. Fit the model

How do we choose the best parameters of our model given our data?

4. Evaluate model performance

How do we evaluate whether this process gave rise to a good model?

Choose a Model

- What is a Model?
- Data 8 Review
 - Regression Line, Correlation
- **The Modeling Process**
 - **Choose a Model**
 - Choose a Loss Function
 - Fit the Model
 - Evaluate the Model

Simple Linear Regression: Our First Model

Simple Linear Regression Model (SLR)

LA
Notation:

$$\hat{y} = a + bx$$

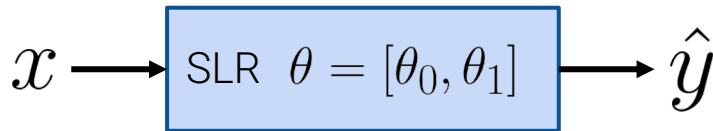
ML

Notation:

$$\hat{y} = \theta_0 + \theta_1 x$$

SLR is a **parametric model**, meaning we choose the “best” **parameters** for slope and intercept based on data.

- We often express θ as a single parameter vector.
- x is **not** a parameter! It is input to our model.



- Note that the true relationship between x and y is usually non-linear. This is why $\hat{y}$ (and not y) appears in our **estimated linear model** expression.

The Modeling Process



1. Choose a model

How should we represent the world?

$$\hat{y} = \theta_0 + \theta_1 x \quad \text{SLR model}$$



2. Choose a loss function

How do we quantify prediction error?

3. Fit the model

How do we choose the best parameters of our model given our data?

$$\hat{y} = \hat{\theta}_0 + \hat{\theta}_1 x$$

4. Evaluate model performance

How do we evaluate whether this process gave rise to a good model?

Reflect

Loss Functions

- What is a Model?
 - Regression Line, Correlation
- **The Modeling Process**
 - Choose a Model
 - **Choose a Loss Function**
 - Fit the Model
 - Evaluate the Model

The Modeling Process



1. Choose a model

How should we represent the world?

$$\hat{y} = \theta_0 + \theta_1 x$$

SLR model

2. Choose a loss function

How do we quantify prediction error?

3. Fit the model

How do we choose the best parameters of our model given our data?

4. Evaluate model performance

How do we evaluate whether this process gave rise to a good model?

Loss Functions

We need some metric of how "good" or "bad" our predictions are.

A **loss function** characterizes the **cost**, error, or fit resulting from a particular choice of model or model parameters.

- Loss quantifies how **bad** a prediction is for a **single** observation.
- If our prediction $\hat{y}$ is **close** to the actual value y , we want **low loss**.
- If our prediction $\hat{y}$ is **far** from the actual value y , we want **high loss**.

$$L(y, \hat{y})$$

There are many definitions of loss functions!

The choice of loss function:

- Affects the accuracy and computational cost of estimation.
- Depends on the estimation task:
 - Are outputs quantitative or qualitative?
 - Do we care about outliers?
 - Are all errors equally costly? (e.g., false negative on cancer test)

Squared Loss

$$L(y, \hat{y}) = (y - \hat{y})^2$$

- Widely used.
- Also called "L2 loss".
- Reasonable:
 - $\hat{y} = y \rightarrow$ good prediction
 $\rightarrow$ good fit $\rightarrow$ no loss
 - $\hat{y}$ far from $y \rightarrow$ bad prediction
 $\rightarrow$ bad fit $\rightarrow$ lots of loss

Absolute Loss

$$L(y, \hat{y}) = |y - \hat{y}|$$

- Sounds worse than it is.
- Also called "L1 loss".
- Reasonable:
 - $\hat{y} = y \rightarrow$ good prediction
 $\rightarrow$ good fit $\rightarrow$ no loss
 - $\hat{y}$ far from $y \rightarrow$ bad prediction
 $\rightarrow$ bad fit $\rightarrow$ some loss

L2 and L1 Loss for SLR

Squared Loss

$$L(y, \hat{y}) = (y - \hat{y})^2$$

- Widely used.
- Also called "L2 loss".
- Reasonable:
 - $\hat{y} = y \rightarrow$ good prediction
 $\rightarrow$ good fit $\rightarrow$ no loss
 - far from $y \rightarrow$ bad prediction
 $\rightarrow$ bad fit $\rightarrow$ lots of loss

For our SLR model $\hat{y} = \theta_0 + \theta_1 x$

$$L(y, \hat{y}) = (y - (\theta_0 + \theta_1 x))^2$$

Absolute Loss

$$L(y, \hat{y}) = |y - \hat{y}|$$

- Sounds worse than it is.
- Also called "L1 loss".
- Reasonable:
 - $\hat{y} = y \rightarrow$ good prediction
 $\rightarrow$ good fit $\rightarrow$ no loss
 - far from $y \rightarrow$ bad prediction
 $\rightarrow$ bad fit $\rightarrow$ some loss

For our SLR model $\hat{y} = \theta_0 + \theta_1 x$

$$L(y, \hat{y}) = |y - (\theta_0 + \theta_1 x)|$$

Question?

Why don't we use residual error directly and instead we use absolute loss or squared loss?

Residuals as loss function?

Why don't we directly use residual error as the loss function?

$$e = (y - \hat{y})$$

- Doesn't work: big negative residuals shouldn't cancel out big positive residuals!
 - Our predictions can be very off, but we can still get a zero residual

Empirical Risk is Average Loss over Data

We care about how bad our model's predictions are for our entire data set, not just for one point.

A natural measure, then, is of the **average loss** (aka **empirical risk**) across all points.

Given data $\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$:

$$\hat{R}(\theta) = \frac{1}{n} \sum_i L(y_i, \hat{y}_i)$$

Function of the parameter θ (holding the data fixed) because θ determines $\hat{y}$.

The average loss on the sample tells us how well the model fits the data (not the population).

But hopefully these are close.

Empirical Risk is Average Loss over Data

We care about how bad our model's predictions are for our entire data set, not just for one point.

A natural measure, then, is of the **average loss** (aka **empirical risk**) across all points.

Given data $\mathcal{D} = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$:

$$\hat{R}(\theta) = \frac{1}{n} \sum_i L(y_i, \hat{y}_i)$$

The colloquial term for average loss depends on which loss function we choose.

L2 loss

**Mean
Squared
Error (MSE)**

$$\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

L1 loss

**Mean
Absolute
Error (MAE)**

$$\frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

The Modeling Process

1. Choose a model



How should we represent the world?

$$\hat{y} = \theta_0 + \theta_1 x$$

SLR model

2. Choose a loss function



How do we quantify prediction error?

$$L(y, \hat{y}) = (y - \hat{y})^2$$

Squared loss

3. Fit the model

How do we choose the best parameters of our model given our data?

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - (\theta_0 + \theta_1 x))^2$$

MSE for SLR

4. Evaluate model performance

How do we evaluate whether this process gave rise to a good model?

The combination of model + loss that we focus on today is known as **least squares regression**.

The Modeling Process

1. Choose a model



How should we represent the world?

$$\hat{y} = \theta_0 + \theta_1 x \quad \text{SLR model}$$

2. Choose a loss function



How do we quantify prediction error?

$$L(y, \hat{y}) = (y - \hat{y})^2 \quad \text{Squared loss}$$

3. Fit the model

How do we choose the best parameters of our model given our data?

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - (\theta_0 + \theta_1 x))^2$$

We want to find $\hat{\theta}_0, \hat{\theta}_1$ that minimize this **objective function**.

4. Evaluate model performance

How do we evaluate whether this process gave rise to a good model?

Fit the Model

- What is a Model?
 - Regression Line, Correlation
- **The Modeling Process**
 - Choose a Model
 - Choose a Loss Function
 - **Fit the Model**
 - Evaluate the Model

Minimizing MSE for the SLR Model

Recall: we wanted to pick the **regression line** $\hat{y} = \hat{\theta}_0 + \hat{\theta}_1 x$

To minimize the (sample) **Mean Squared Error**: $MSE(\theta_0, \theta_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (\theta_0 + \theta_1 x_i))^2$

To find the best values, we set derivatives equal to zero to **obtain the optimality conditions**:

$$\frac{\partial}{\partial \theta_0} MSE = 0$$

$$\frac{\partial}{\partial \theta_1} MSE = 0$$

Partial Derivative of MSE with Respect to θ_0, θ_1

$$\frac{\partial}{\partial \theta_0} MSE = \frac{\partial}{\partial \theta_0} \frac{1}{n} \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i)^2$$

Derivative of sum is sum of derivatives

$$= \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta_0} (y_i - \theta_0 - \theta_1 x_i)^2$$

Chain rule

$$= \frac{1}{n} \sum_{i=1}^n 2(y_i - \theta_0 - \theta_1 x_i)(-1)$$

Simplify constants

$$= -\frac{2}{n} \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i)$$

$$\frac{\partial}{\partial \theta_1} MSE = \frac{\partial}{\partial \theta_1} \frac{1}{n} \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i)^2$$

Derivative of sum is sum of derivatives

$$= \frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta_1} (y_i - \theta_0 - \theta_1 x_i)^2$$

Chain rule

$$= \frac{1}{n} \sum_{i=1}^n 2(y_i - \theta_0 - \theta_1 x_i)(-x_i)$$

Simplify constants

$$= -\frac{2}{n} \sum_{i=1}^n (y_i - \theta_0 - \theta_1 x_i) x_i$$

Estimating Equations

To find the best values, we set derivatives equal to zero to **obtain the optimality conditions**:

$$\begin{aligned} 0 = \frac{\partial}{\partial \theta_0} MSE &= -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{\theta}_0 - \hat{\theta}_1 x_i) \iff \frac{1}{n} \sum_i y_i - \hat{y}_i = 0 \\ 0 = \frac{\partial}{\partial \theta_1} MSE &= -\frac{2}{n} \sum_{i=1}^n (y_i - \hat{\theta}_0 - \hat{\theta}_1 x_i) x_i \iff \frac{1}{n} \sum_i (y_i - \hat{y}_i) x_i = 0 \end{aligned}$$

“Equivalent”

Estimating equations

To find the best θ_0, θ_1 , we need to solve the **estimating equations** on the right.

From Estimating Equations to Estimators

Goal: Choose θ_0, θ_1 to solve two estimating equations:

$$\frac{1}{n} \sum_i y_i - \hat{y}_i = 0 \quad \boxed{1} \quad \text{and} \quad \frac{1}{n} \sum_i (y_i - \hat{y}_i) x_i = 0 \quad \boxed{2}$$

1

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (y_i - \hat{\theta}_0 - \hat{\theta}_1 x_i) &= 0 \iff \overbrace{\left(\frac{1}{n} \sum_{i=1}^n y_i \right)}^{\bar{y}} - \hat{\theta}_0 - \hat{\theta}_1 \overbrace{\left(\frac{1}{n} \sum_{i=1}^n x_i \right)}^{\bar{x}} = 0 \\ &\iff \bar{y} - \hat{\theta}_0 - \hat{\theta}_1 \bar{x} = 0 \\ &\iff \hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x} \end{aligned}$$

From Estimating Equations to Estimators

Goal: Choose θ_0, θ_1 to solve two estimating equations:

$$\frac{1}{n} \sum_i y_i - \hat{y}_i = 0 \quad \boxed{1} \quad \text{and}$$

$$\frac{1}{n} \sum_i (y_i - \hat{y}_i) x_i = 0 \quad \boxed{2}$$

Now, let's try: $\boxed{2} - \boxed{1} * \bar{x}$

$$\frac{1}{n} \sum_i (y_i - \hat{y}_i) x_i - \frac{1}{n} \sum_i (y_i - \hat{y}_i) \bar{x} = 0 \quad \Longleftrightarrow \quad \frac{1}{n} \sum_i (y_i - \hat{y}_i) (x_i - \bar{x}) = 0$$

$$\left(\text{using } \hat{y}_i = \hat{\theta}_0 + \hat{\theta}_1 x_i \right) \Rightarrow \frac{1}{n} \sum_{i=1}^n \left(y_i - \hat{\theta}_0 - \hat{\theta}_1 x_i \right) (x_i - \bar{x}) = 0$$

$$\left(\text{using } \hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x} \right) \Rightarrow \frac{1}{n} \sum_i \left(y_i - \bar{y} + \hat{\theta}_1 \bar{x} - \hat{\theta}_1 x_i \right) (x_i - \bar{x}) = 0$$

$$\Rightarrow \frac{1}{n} \sum_i \left((y_i - \bar{y}) - \hat{\theta}_1 (x_i - \bar{x}) \right) (x_i - \bar{x}) = 0$$

From Estimating Equations to Estimators

$$\Rightarrow \frac{1}{n} \sum_i \left[(y_i - \bar{y})(x_i - \bar{x}) - \hat{\theta}_1 (x_i - \bar{x})^2 \right] = 0$$

$$\Rightarrow \frac{1}{n} \sum_i (y_i - \bar{y})(x_i - \bar{x}) = \hat{\theta}_1 \frac{1}{n} \sum_i (x_i - \bar{x})^2$$

Plug in definitions of correlation and SD:

$$r \sigma_y \sigma_x = \hat{\theta}_1 \sigma_x^2$$

Solve for $\hat{\theta}_1$:

$$\hat{\theta}_1 = r \frac{\sigma_y}{\sigma_x}$$

Reminder

$$\sigma_x^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$$

$$r = \frac{1}{n} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sigma_x} \right) \left(\frac{y_i - \bar{y}}{\sigma_y} \right)$$

Estimating Equations

Estimating equations are the equations that the model fit has to solve. They help us:

- Derive the estimates
- Understand what our model is paying attention to

$$\frac{1}{n} \sum_i y_i - \hat{y}_i = 0$$

$$\frac{1}{n} \sum_i (y_i - \hat{y}_i) x_i = 0$$

For SLR:

- The residuals should **average to zero** (otherwise we should fix the intercept!)
- The residuals should be **orthogonal to the predictor variable** (or we should fix the slope!)

Very important for HW 5

What is a Model?

- **What is a Model?**
 - Regression Line, Correlation
- The Modeling Process
 - Choose a Model
 - Choose a Loss Function
 - Fit the Model
 - Evaluate the Model

The Modeling Process

1. Choose a model



How should we represent the world?

$$\hat{y} = \theta_0 + \theta_1 x \quad \text{SLR model}$$

2. Choose a loss function



How do we quantify prediction error?

$$L(y, \hat{y}) = (y - \hat{y})^2 \quad \text{Squared loss}$$

3. Fit the model



How do we choose the best parameters of our model given our data?

$$\hat{R}(\theta) = \frac{1}{n} \sum_{i=1}^n (y_i - (\theta_0 + \theta_1 x))^2 \quad \text{MSE for SLR}$$

4. Evaluate model performance

How do we evaluate whether this process gave rise to a good model?

$$\hat{y} = \hat{\theta}_0 + \hat{\theta}_1 x \quad \left\{ \begin{array}{l} \hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x} \\ \hat{\theta}_1 = r \frac{\sigma_y}{\sigma_x} \end{array} \right.$$

Next Time!